

1U CubeSat Comms Board — 437 MHz BPSK Downlink, 145.9 MHz FM-AFSK Uplink, RP2040-Controlled

Applications

- centisat 1U–3U CubeSat platform (MSU senior capstone)
- University / educational CubeSat narrow-band telemetry and command
- AMSAT-style amateur-satellite VHF/UHF inverse-band comms

Features

Topology

- 70 cm BPSK downlink / 2 m FM-AFSK uplink**, half-duplex
- Single Si5351A drives TX carrier (CLK0) and RX local oscillator (CLK1)
- RP2040 (Pico module) handles mod/demod and IHU SPI link
- Separate TX/RX SMA antennas (no diplexer)

Electrical

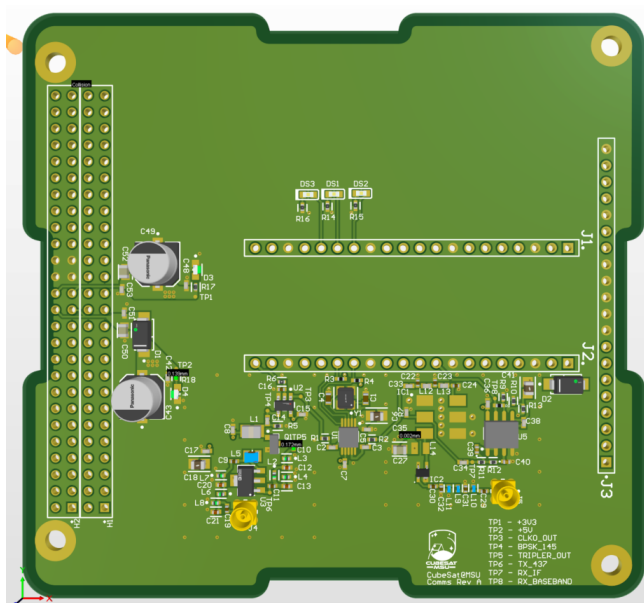
- TX:** 437 MHz BPSK, **+13 dBm** typ output
- TX chain: 74LVC1G86 XOR mod (145.67 MHz) → BFR92A tripler → 3-pole BPF → ADL5602 MMIC (+20 dB) → output BPF
- RX:** 145.9 MHz FM-AFSK (1200/2200 Hz), **~1 dB system NF**
- RX chain: 2m BPF → PSA4-5043+ LNA → ADE-1+ passive mixer → Sallen-Key LPF (3.3 kHz) → MCP6022 → RP2040 ADC
- 5-pole LC LPF on LO line suppresses LO 3rd-harmonic at 437 MHz (~34 dB)

SWaP

- CSKB stack-bus form factor (~96 × 90 mm), single board
- Mass < 80 g (estimated)
- +3 V3 ~150 mA + +5 V ~140 mA ≈ 1.2 W** (RX active)

Environment

- Operating temperature **–20 °C to +60 °C** (preliminary)
- 4-layer FR-4 (JLCPCB JLC04161H-7628), 1.6 mm
- 50 Ω microstrip (0.358 mm) on L1 referenced to L2 GND plane



Project Information

Field	Value
Project	centisat Comms
Schematic rev	1.3
Board rev	Rev A (in layout)
Repository	github.com/ngrabbs/centisat
Owner	MSU CubeSat (capstone)

BRING-UP STATUS

Schematic complete (Rev 1.3); 4-layer PCB in layout. Si5351A clock path verified on Adafruit breakout + Pico 2. RX front-end (PSA4 LNA + ADE-1+ mixer) replaces original SA612 due to EOL — see design/rx_mixer_trade_study.md. **Not flight-qualified.**

Protection & Safety

- Reverse-polarity D4 + VSYS isolation D5 on +5 V / Pico inputs
- 2 m BPF protects LNA/mixer from 437 MHz self-TX (>40 dB)
- 5-pole LO LPF suppresses LO harmonics into the mixer
- 51 Ω IF port termination on ADE-1+
- Si5351A outputs disabled at boot — TX off by default

Interface

- CSKB H1/H2 stack-bus (2× Samtec ESQ-126-39-G-D)

- SPI0 (slave) + I²C0 + IRQ to IHU via CSKB H1
- TX (J4) and RX (J5) on **MMCX in-board** connectors
- SWD debug header (J2, optional)